Vaidehi Som

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Education

University of Pennsylvania | Masters Robotics Engineering (Specializing in CV & DL) Aug'22 – May'24
Indian Institute of Technology (IIT) Jammu, India | Bachelors Mechanical Engineering Aug'17 – Jun'21

Work Experience

Perception Navigation Engineer | *Rust, Deep Learning, Computer Vision, SLAM, Sensor Fusion* May'23-Dec'23, May'24-*
Zipline International Inc. (Drone delivery & Logistics), USA

- Improving feature matching using learned features. Realtime inferencing as onnx model and rust model. Evaluation metrics to compare classical and DL model
- Implemented **incremental adaptive graph batch optimization** in the VIO pipeline serving as ground truth for real-world datasets
- **Sensor Fusion**: Implemented fusion of GNSS and VIO in **factor graph** in a **tightly coupled** fashion

Mobile Robotics Software Engineer | *C++, ROS, Startup, Automated Guided Vehicle, Docker* Aug'21 – Jun'22
Addverb Technologies (Warehouse Automation), India

- Implemented communication protocols(**serial/CAN, UART**) for AMR's **LiDAR** and **IMU** for low-level drivers
- Pipeline creation: Implemented safety relevant **Pure pursuit, Lyapunov(research paper) controller** and safety packages for **navigation** stack using **C++** and **ROS**. Improved odometry with calibration, controllers, and **IMU** using **Kalman filter**
- Achieved a **50%** reduction in testing time through the automation of odometry calibration and sensor testing processes

Projects

Stereo Visual Odometry | *Autonomous Driving, Multi-threading, SLAM, Geometric Computer Vision, C++* *Results/Code*

- Implemented **Visual SLAM pipeline** for stereo images to find 3D locations of keypoints, used **GFTT** for feature detection and **triangulation** for 3D point location. Implemented **direct method** and **optical flow** for pose and feature estimation during feature tracking and **Bundle Adjustment** for backend optimization using **g2o**

Dense 3D reconstruction from point clouds using Octrees | *C++, PointCloud, Octree, SLAM* *Results/Code*

- Implemented **octomap::OcTree** for efficient SLAM mapping, opting for a basic octree structure over RGB-enhanced variants to prioritize occupancy information and simplify map construction
- Employed Octomap's **point cloud** to translate RGB-D and camera pose data into world coordinates, updating the octree map's occupancy probabilities, and compressed the final map into an octomap.bt file for visualization with **octovis**

Vision based SLAM

- Backend- **Bundle adjustment** with **ceres** with BAL dataset (C++) *Code*
- Implemented **2-view** and **multiple view stereo** algorithms to convert multiple 2D viewpoints into 3D reconstruction *Code*
- Recovering 3D transformation between two views using **RANSAC**, Pose recovery and 3D reconstruction
- Augmented Reality with AprilTags using both **PnP and P3P algorithm** *Code*

Semantic Segmentation using SegFormers | *Python, Transformers, ResNet-50* *Results/Code*

- Implemented Segformers, integrating Efficient Self-Attention, Mix-FFN, Overlap Patch embedding, and Dice loss
- Used the 3k images from imbalanced BDD 100K dataset to achieve a Mean Intersection over Union (IoU) score of 77.07%

Multi Object Detection and Tracking | *Python, DeepSORT, YOLO8* *Results/Code*

- Developed a Multiple Object Tracking System with **DeepSORT** and **YOLOv8**, employed Kalman filtering and the **Hungarian** algorithm for improved data association. Deployment done on NVIDIA Triton Inference Server via Docker optimizing dynamic batching. Converted the model to **TensorRT**, achieving increased throughput and reduced latency on **NVIDIA Jetson Orin**.

Neural Network optimization using Quantization and Pruning | *Python, Deep Learning, KD loss* *Results/Code*

- Quantized CNN-based architectures to make them **4x** smaller and **2x** faster in Pytorch
- Set up **Knowledge Distillation** to reduce inference time by **12x** and affecting results only by **13%** on segmentation pipeline

Research Publications

LIV: Language-Image Representations and Rewards for Robotic Control | *Multimodal learning* *Paper*
*Yecheng Jason Ma, **Vaidehi Som***, William Liang*, Vikash Kumar, Amy Zhang, Osbert Bastani, Dinesh Jayaraman* *ICML 2023*

Human-Robot Co-Learning and Feedback Insights using Sequential Transfer Learning
Sequential Transfer Learning for human decision making model during Human Robot CoLearning

Technical Skills

Programming: C++, Python, Rust, CUDA, Linux, CMake, Git, Docker, VIM, gdb, Git/Github, GTest, Jetson Nano
Robotics: ROS(1&2), OpenCV, Eigen, Sensor Fusion, Ceres, g2o, Sensor Synchronization, Optimization (LM, GN), SLAM, GTSAM
AI/ML: PyTorch, Pandas, Numpy, ML Ops, GPU, scikit-learn, Scipy, Matplotlib, Weights&Biases, TensorRT, Segmentation